The order-nine geometry from pairs and matchings

Human Compression Project

Abstract

The geometry with two S_4 parabolics in $\mathrm{PSL}(2,9)$ has a concrete model on the pairs and perfect matchings of a six-element set. This model identifies it with the symplectic generalized quadrangle of order two. An explicit duality and Tutte's stabilizer bound determine its full automorphism group and exact arc transitivity. We recover the order-nine assertions of the fixed Part II source without computing a graph automorphism group. The classical isomorphism $\mathrm{PSL}(2,9) \cong A_6$, its subgroup classes, its automorphism group, and Tutte's theorem are stated premises.

1 A concrete incidence geometry

Let $\Omega = \{1,2,3,4,5,6\}$. A *duad* is a two-element subset of Ω; a *syntheme* is a partition of Ω into three duads. Let X have the fifteen duads and the fifteen synthemes as its two vertex types. A duad is incident with a syntheme when it is one of its parts. Every vertex has degree three. The natural S_6 action is faithful, since a permutation fixing every two-element subset fixes their singleton intersections.

Lemma 1. *This incidence geometry is the symplectic generalized quadrangle of order* $(2,2)$*. Its incidence graph is connected, has girth eight and has diameter four.*

Proof. For a duad d not in a syntheme L, its two letters occur in two different parts of L. Exactly one part e of L is disjoint from d. There is a unique syntheme containing d and e: its third part consists of the two unused letters. Thus there is a unique point of L collinear with d. Two duads are on at most one syntheme, and two synthemes share at most one duad. These facts are precisely the generalized quadrangle axioms, with three incidences at each vertex.

For completeness, its symplectic realization is equally explicit. Let $W \subset \mathbb{F}_2^6$ consist of the vectors with even coordinate sum, and put $V = W/\langle(1,1,1,1,1,1)\rangle$. The ordinary dot product restricts to an alternating form on W; its radical is the displayed one-dimensional subspace. It therefore induces a nondegenerate alternating form on the four-dimensional space V. Every nonzero class has a unique representative of weight two, so its fifteen projective points are precisely the duads. Distinct duads are orthogonal exactly when they are disjoint. The sum of two disjoint duads is represented by their complementary duad, since adding the all-ones vector replaces a set by its complement. Consequently the totally isotropic two-dimensional subspaces are exactly the synthemes. This proves the claimed realization.

The quadrangle axiom gives distance three for a nonincident point and line, and distance at most four between any two vertices. Two intersecting duads are not collinear, so their distance is four. There is no four-cycle because two points have at most one common line, and no six-cycle by the uniqueness in the quadrangle axiom. There is an eight-cycle: the successive duads $12, 34, 15, 36$ are joined by the four synthemes $12|34|56$, $15|26|34$, $15|24|36$, $12|36|45$. Thus the diameter and girth are exactly as asserted. □

Here ab denotes $\{a, b\}$. The dot-product construction also identifies the type-preserving S_6 action with $\mathrm{Sp}_4(2)$. Indeed it embeds S_6 faithfully in that group, whose order is $15 \cdot 8 \cdot 3 \cdot 2 = 720$: choose the first nonzero symplectic basis vector, its partner, and then a symplectic basis of their two-dimensional orthogonal complement.

2 The specified simple group and its configuration

Put $G = A_6$. We use the classical isomorphism $G \cong \mathrm{PSL}(2, 9)$ and the two maximal S_4 classes in this group. These are precisely the two classes supplied by the subfield $\mathrm{PGL}(2, 3)$ case of the Dickson subgroup theorem; they are not conclusions of the geometry classification being audited.

Lemma 2. *The duad and syntheme stabilizers in G represent those two S_4 classes. An incident pair has intersection D_8. The action on each local set of three neighbors is two-transitive. Every configuration of two distinct S_4 subgroups meeting in D_8 is conjugate to one such incident pair, under the unordered convention.*

Proof. The duad stabilizer is $(S_2 \times S_4) \cap A_6$, isomorphic to S_4 by its action on the other four letters: parity uniquely determines the action on the duad. The syntheme stabilizer is the even part of $C_2 \wr S_3$, namely $C_2^2 \rtimes S_3 \cong S_4$. It acts transitively on the six letters, whereas the duad stabilizer has orbits of lengths two and four. They cannot be conjugate even in S_6, and hence represent the two given G-classes. Both classes have fifteen members, since both actions are transitive.

The duad stabilizer acts as the full S_3 on the three matchings of the remaining four letters; the syntheme stabilizer acts as the full S_3 on its three parts. Thus both local actions are two-transitive. An incident-pair stabilizer has order $24/3 = 8$ and is a Sylow two-subgroup of S_4, hence D_8. Denote it by B.

Its orbits on the six letters have sizes two and four. A normalizer of B in A_6 must preserve the orbit of size two and therefore belongs to the duad stabilizer. A Sylow two-subgroup of S_4 is self-normalizing, so $N_G(B) = B$. There are therefore $360/8 = 45$ conjugate Sylow two-subgroups of G. Each S_4 contains three of them. For each of the two S_4 classes, counting incidences gives $15 \cdot 3 = 45$; transitivity on the Sylow subgroups shows that each one belongs to exactly one S_4 of each class. It follows that every eligible pair is conjugate to the displayed pair. Its coset incidence graph is X: the natural transitive actions identify both coset sets, and flag transitivity identifies their incidence relation. □

3 An explicit duality and the full group

The following table gives a permutation d of the thirty vertices. Each row is a transposition between the duad and the syntheme shown. In particular $d^2 = 1$ and d interchanges the two types.

Duad	Syntheme	Duad	Syntheme	Duad	Syntheme
12	12|34|56	13	14|25|36	14	13|26|45
15	16|24|35	16	15|23|46	23	16|23|45
24	15|24|36	25	13|25|46	26	14|26|35
34	12|35|46	35	15|26|34	36	13|24|56
45	14|23|56	46	16|25|34	56	12|36|45

For every incidence $a \in L$, the syntheme $d(a)$ contains the duad $d(L)$. This follows by substituting the three parts of each of the fifteen synthemes in the table. Thus d is a graph automorphism. The accompanying literal reader checks exactly these forty-five incidences, the permutation property and the reversal of types. The witness was found by a graph-isomorphism routine; its verification uses neither that routine nor a group-order computation.

Theorem 3. *The full automorphism group of X is isomorphic to* $\mathrm{P\Gamma L}(2,9)$ *and has order* 1440*. It is five-arc-transitive and is not six-arc-transitive.*

Proof. Let $H = \mathrm{Aut}(X)$. It contains the faithful type-preserving S_6 and the type-reversing d, so $|H| \geq 2 \cdot 720 = 1440$. The action is vertex transitive; the local actions in Lemma 2 make it arc transitive. Tutte's theorem for finite connected simple cubic arc-transitive graphs gives $|H_x| \leq 48$. Since X has thirty vertices, $|H| \leq 30 \cdot 48 = 1440$. Equality holds.

To identify the group, its subgroup $G = A_6$ has index four. The action of G on the four cosets H/G is a homomorphism to S_4. Simplicity and $|G| = 360 > 24$ force this homomorphism to be trivial. Hence G lies in its core in H and is normal in H. The two G-orbits on vertices have nonconjugate stabilizers by Lemma 2, so a permutation centralizing G cannot interchange them. Within either orbit, its centralizer is $N_G(M)/M = 1$, since the maximal S_4 stabilizer is self-normalizing in the simple group. Thus $C_H(G) = 1$, and conjugation embeds H in $\mathrm{Aut}(G)$. The classical automorphism theorem identifies $\mathrm{Aut}(\mathrm{PSL}(2,9))$ with $\mathrm{P\Gamma L}(2,9)$, also of order 1440; the embedding is therefore an isomorphism.

Tutte's theorem further says that the full group is regular on s-arcs for some $s \leq 5$. The number from a fixed vertex is $3 \cdot 2^{s-1}$. Since $|H_x| = 48$, necessarily $s = 5$. At length six there are 96 such arcs, exceeding the stabilizer order, so six-arc transitivity is impossible. □

4 Scope and attribution

This proves the $q = 9$ row with two S_4 parabolics in Table 6 and the generalized-quadrangle and full-group assertions on p. 386 of the fixed Part II source. The specified G-action has local arc length four by the separately accepted stabilizer argument; the full group has length five as proved here. The classical duad–syntheme model, symplectic realization and Tutte results are reused, not claimed as new discoveries. Our contribution here is the complete derivation and explicit witness within this source audit. The other source families remain separate; this component adds no whole-original completion, new conjecture answer, Lean verification or compression measurement.

References

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[3] W. T. Tutte, *On the symmetry of cubic graphs*, Canadian J. Math. **11** (1959), 621–624.